

# Appendices

Habitat heterogeneity and phenotypic variation: Site temperature largely predicts diverse spawning portfolios of an interior Chinook Salmon stock

28 April 2026

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## A1 Datasets

### A1.1 Spawn timing data

Spawn timing data for Chinook salmon were collected from 2001 to 2005 in the MFSR. Because redds were not observed daily, we inferred spawn dates as the initial date (day of year) a completed redd was observed. We removed data from 2001, and data from Knapp Creek and Cape Horn Creek, as these sites were not consistently sampled. The final dataset contained 3016 georeferenced redds (i.e., inferred spawn date) across 104 stream reaches (i.e., COMIDs), nested within six major spawning tributaries that also included Beaver Creek (tributary to Marsh Creek) and Elk Creek (tributary to Bear Valley Creek), totaling eight spawning streams across four years. We spatially joined each redd GPS location to the NHDPlus Version 2 to assign stream reaches based on a common identifier (COMID). The COMID is used to link redd data with covariate data associated with the stream reach on which it is located.

#### Grouping Structure

The structure of the data is repeated measures on COMIDs, and multiple years, shared across COMIDs within streams. Many COMIDs only have 1-2 observation (Figure A1.1). There are 23 COMIDs with  $<5$  redds (26%), 13 with  $\leq 2$ . (12.5%).

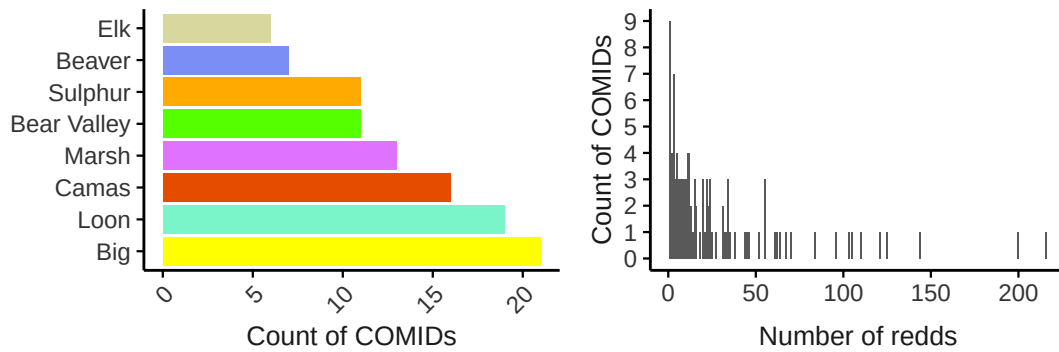


Figure A1.1: Number of observations (redds) per COMID.

## Distribution of spawn timing

Spawn timing is generally unimodal, with a peak in late August to early September (@refig-spawn-dist A). The distribution is slightly left-skewed, but the mean and median are similar, indicating low skewness, and the variance is low with no evidence of overdispersion. There is also substantial variation in spawn date by stream and year (Figure A1.2 B-C; Figure A1.3) and within COMIDs within streams (Figure A1.4).

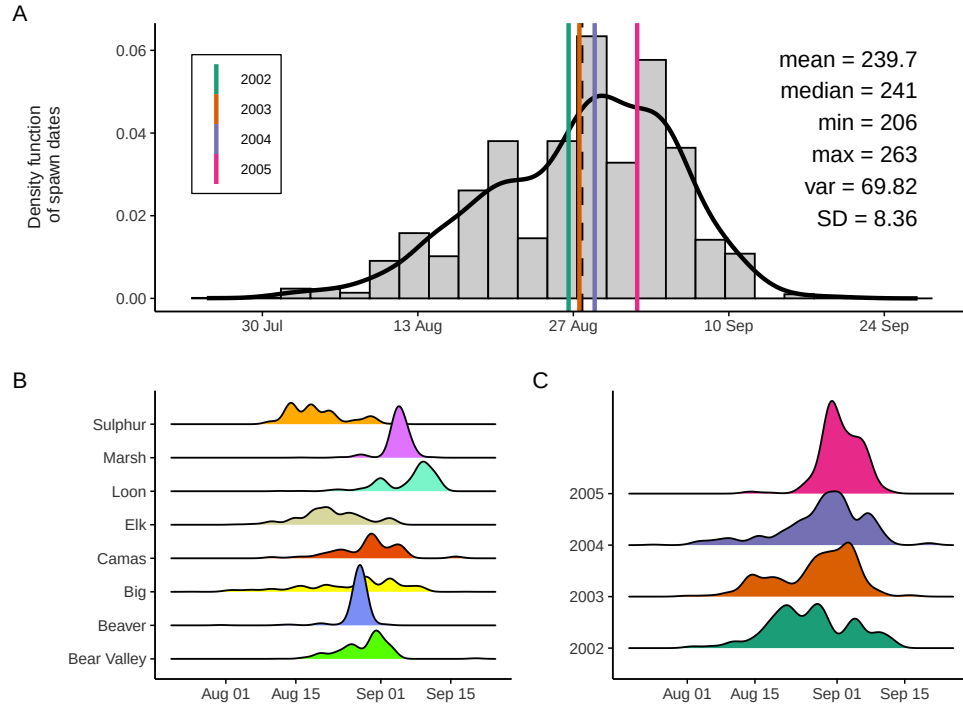


Figure A1.2: Histogram and density of spawn timing for all streams and years (A), and by stream (B) and year (C). In panel (A), colored, vertical lines indicate the mean spawn date for each year., and the black vertical line indicates the global mean.

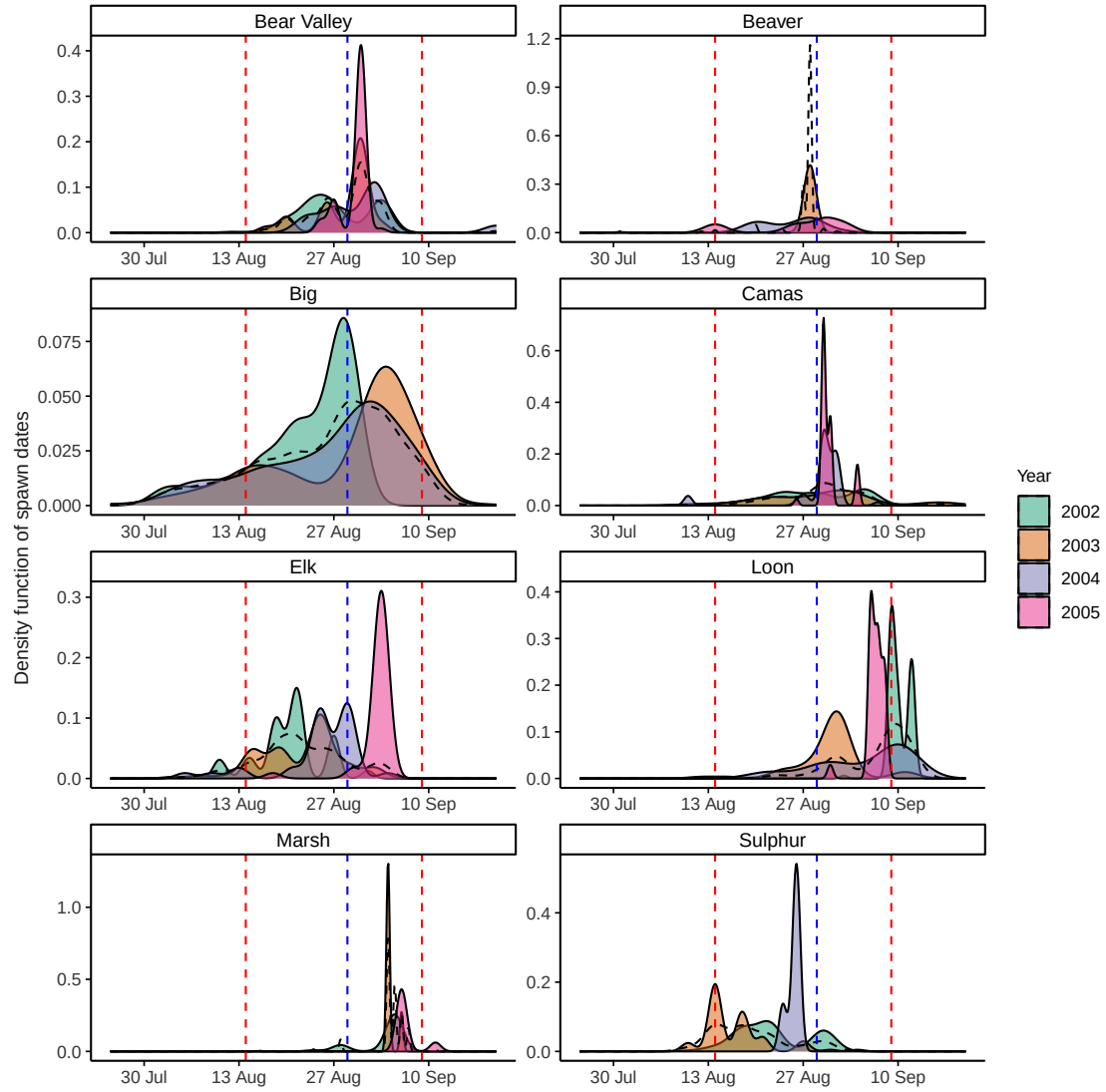


Figure A1.3: Kernel density plots showing the overall distribution of spawn dates by stream and year.

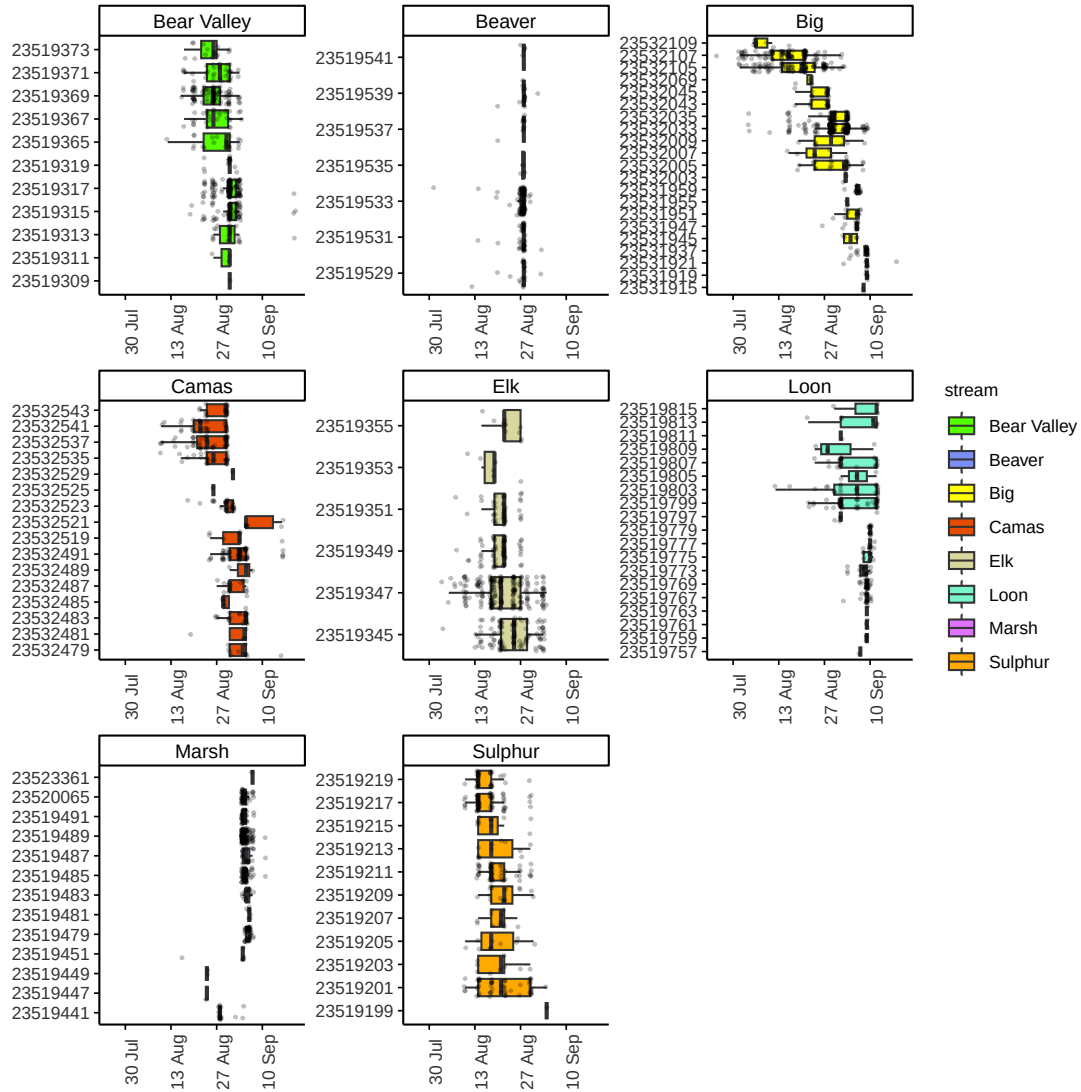


Figure A1.4: Variation in Chinooks salmon spawn timing by COMID within streams.

## Cumulative proportion of redds over time

The temporal progression of Chinook salmon spawning activity varied within each stream across the four study years (Figure A1.5). The cumulative proportion of redds provides an intuitive measure of the spawning season's pace and timing. Streams like Marsh and Sulphur exhibit rapid increases, suggesting short, concentrated spawning windows. In contrast, streams like Big and Camas show more gradual increases, indicating a more protracted spawning season. Year-to-year variation is evident in several streams. Overall, this figure highlights both spatial and temporal heterogeneity in spawn timing that underpins the need for models incorporating stream- and year-specific effects.

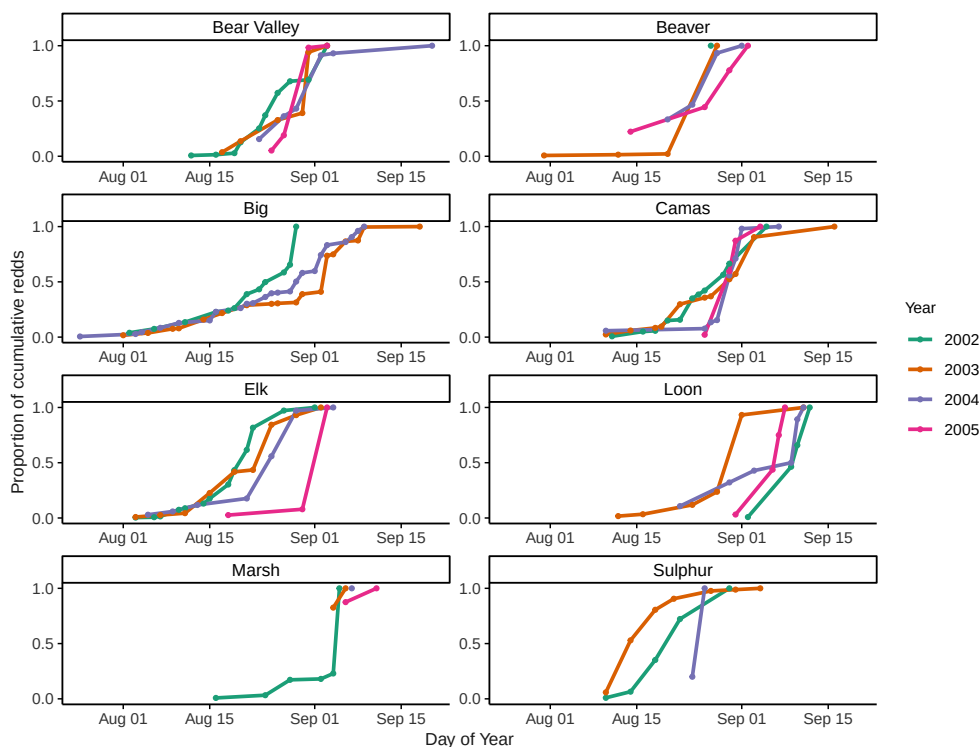


Figure A1.5: Proportion of cumulative Chinook salmon redds over time (day of year) across years (2002–2005) and streams. Each line represents a different year, with color denoting the year. Stream-specific panels illustrate temporal variation in the progression of spawning activity, as measured by cumulative redd counts normalized to the maximum value in each stream-year combination.

## A1.2 Covariate Derivation and Screening

The initial focal covariates included stream temperature ( $^{\circ}\text{C}$ ), stream discharge (cms), elevation (m), and stream slope (unitless).

### Stream temperature

We used modeled daily average stream temperature values predicted at the stream segment (COMID) scale (Siegel et al. 2023). These data were downloaded and filtered to 2002-2005 and for the MFSR (?@fig-plot-temp-regime).

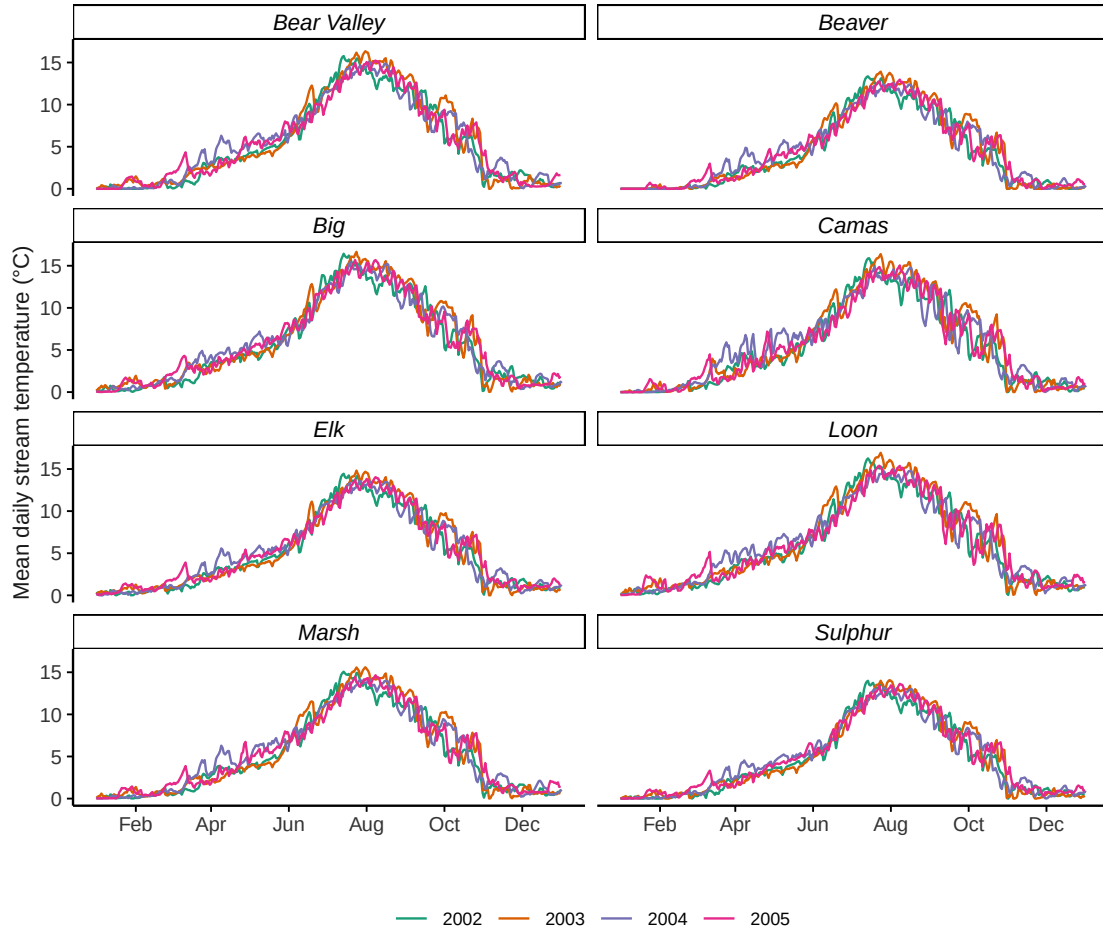


Figure A1.6: Modeled thermal regimes (2001-2005) for MFSR tributaries.

These modeled temperatures were validated against empirical logger data collected at a subset of sites in the Middle Fork basin, with strong agreement (Pearson correlation coefficient = 0.97; Figure A1.7).

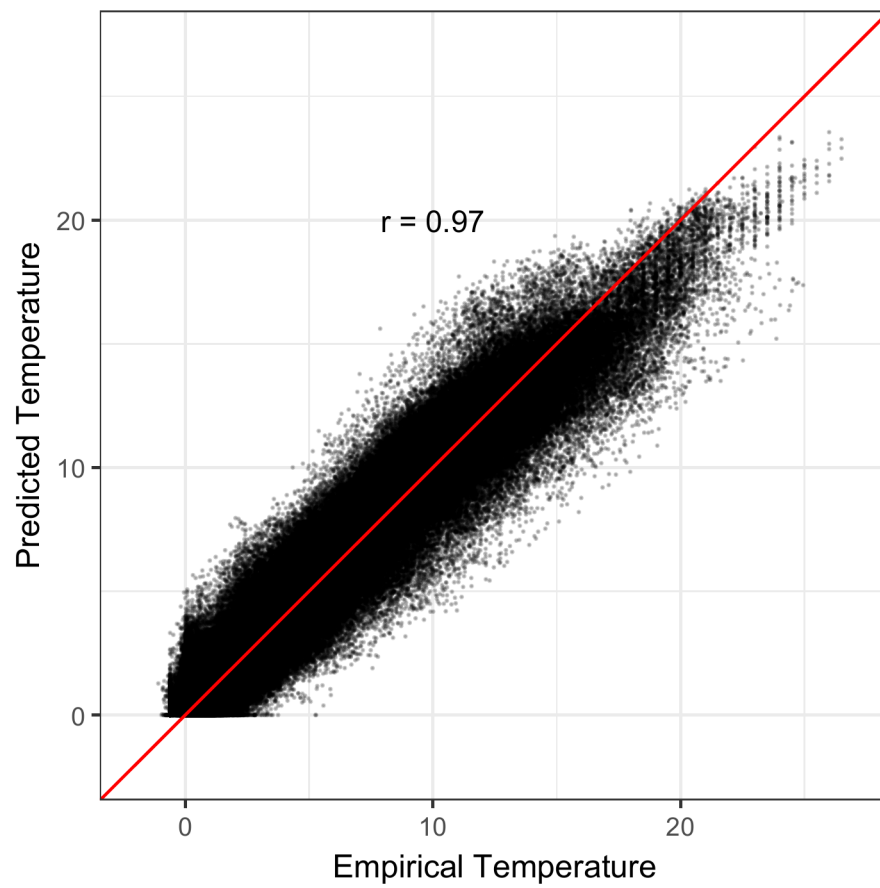


Figure A1.7: Comparison of Siegel et al. (2023) modeled temperatures with empirical observations from Isaak et al.



## Streamflow

Stream flow data were compiled from a single USGS Gage lower in the watershed (MF Salmon River at MF Lodge NR Yellow Pine ID - 13309220; Figure A1.8).

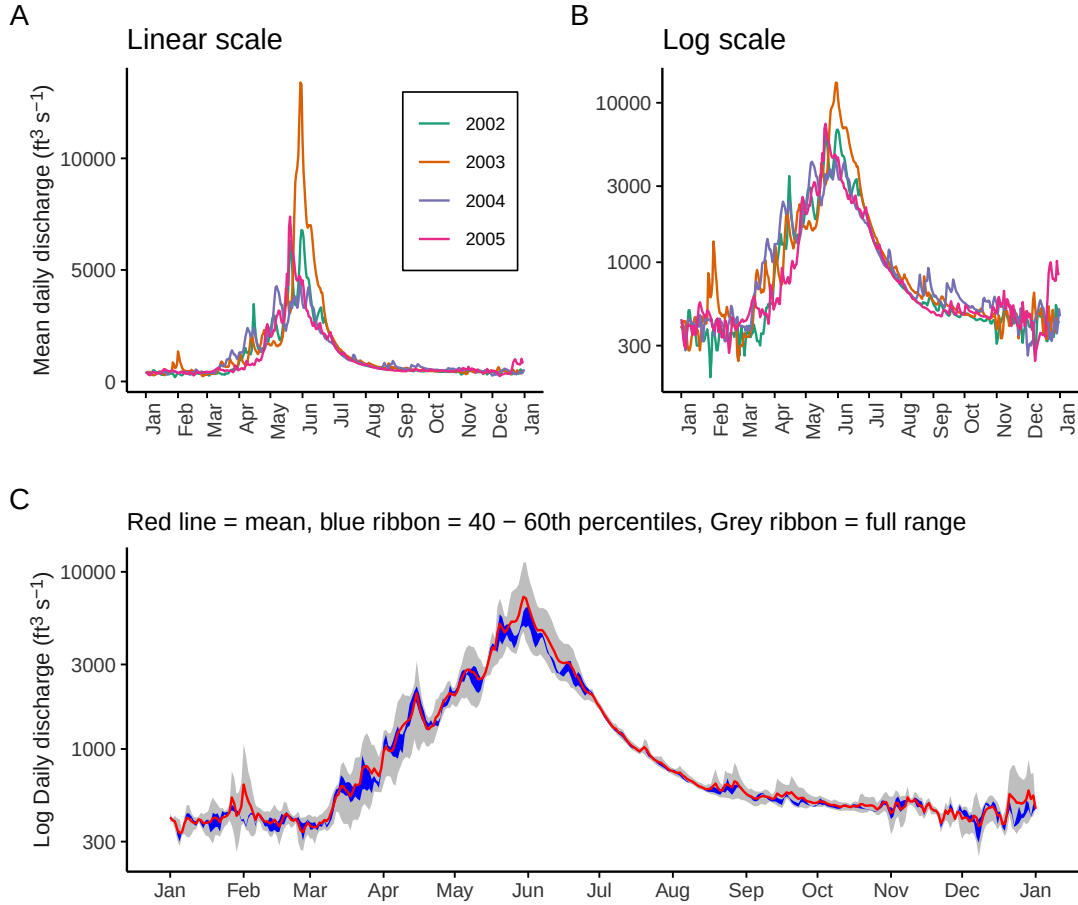


Figure A1.8: Inter-annual variability in daily discharge (cfs) at MF Lodge USGS Gage 13309220.

## Time-windowed temp and flow metrics

For each redd, we computed several time-windowed summaries of temperature and flow relative to the inferred spawn date:

- *Antecedent metrics*: average 30-, 60-, and 90-days preceding the spawn date
- *Spanning metrics*: averages centered on the spawn date
- *Post-spawn metrics*: average 30-, 60-, and 90-day following the spawn date

The spanning and post-spawn metrics were omitted from further consideration as preliminary data exploration showed weak if any relationship with spawn timing.

## Elevation and slope

Elevation and stream slope data were available at the COMID (stream reach) scale from the NHDPlus Version 2 (Horizon Systems, 2018).

## A2 Exploratory Data Analysis

### A2.1 Bivariate relationships

Bivariate relationships between spawn date (`yday`) and continuous predictors (Figure A2.1)), and pairwise correlations between all continuous covariates (Figure A2.2), are shown below.

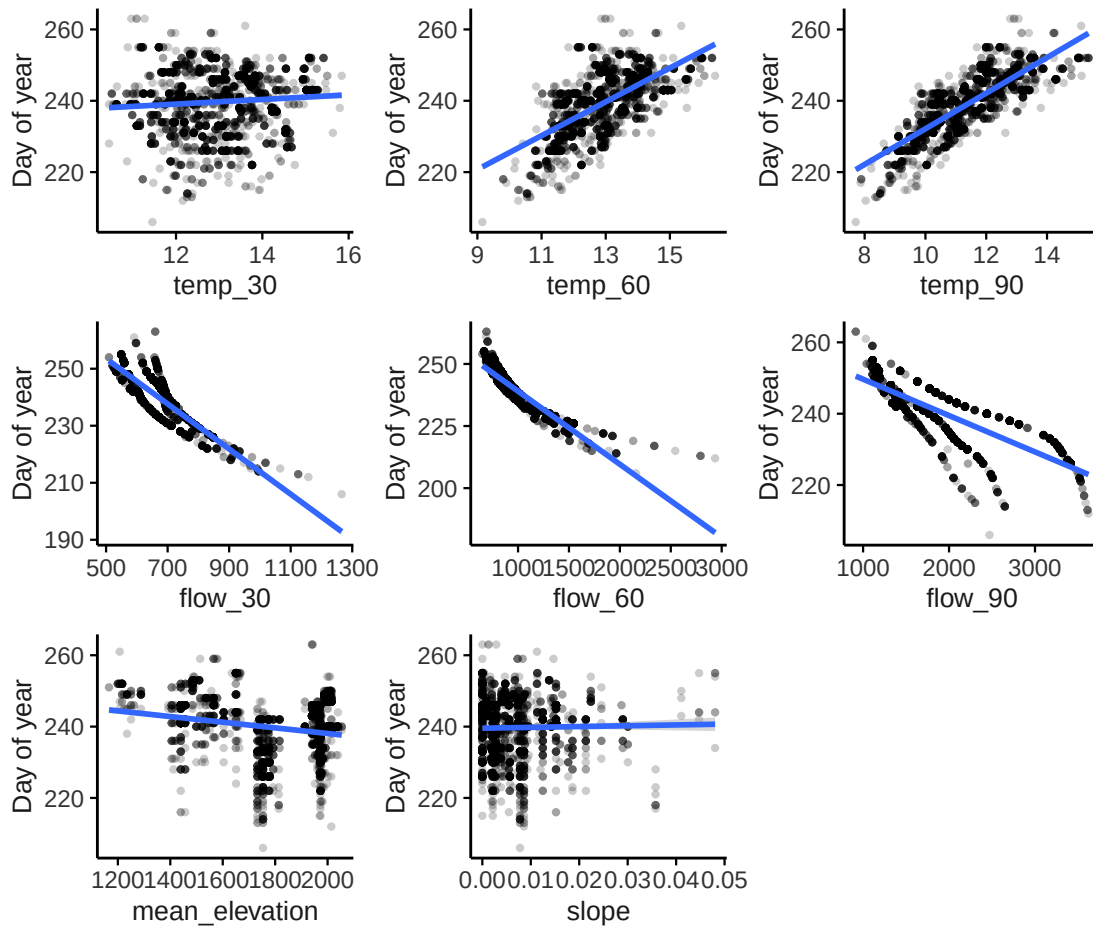


Figure A2.1: Bivariate relationships between spawn date (`yday`) and continuous covariates. Solid lines are linear fits.

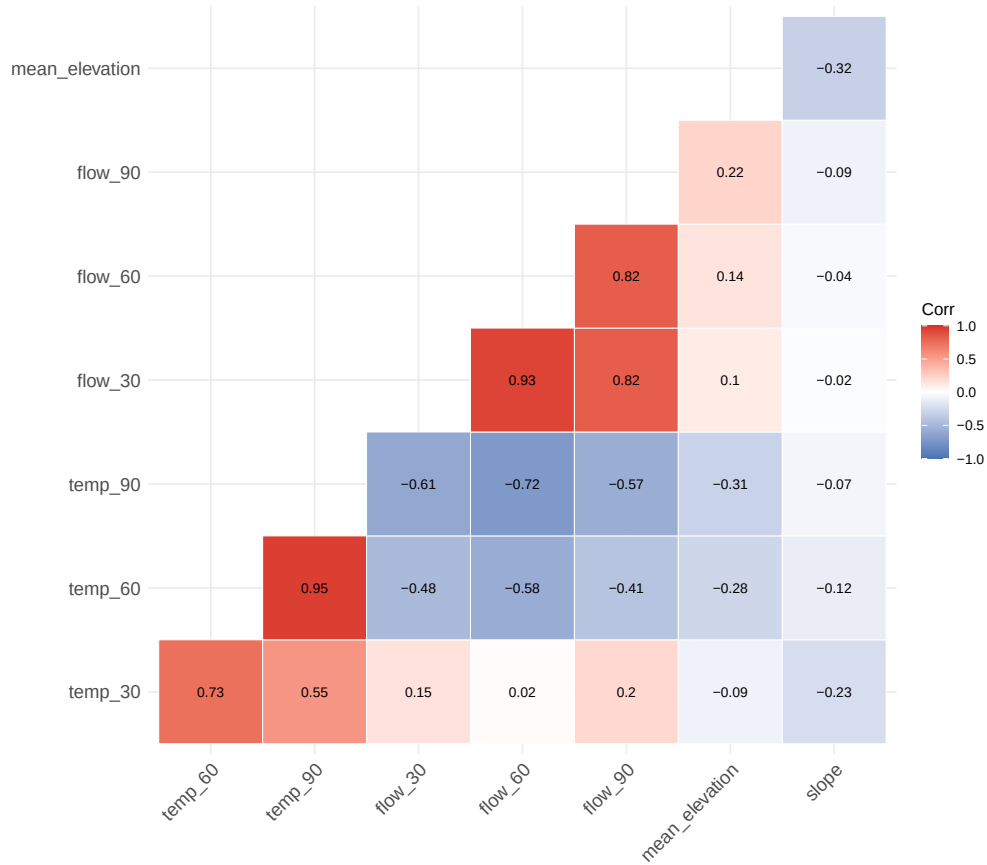


Figure A2.2: Pairwise correlations between continuous covariates.

### Takeaways from Figures

Among temperature variables, **temp\_90** clearly shows the strongest relationship with spawn date (Figure A2.1) and is highly colinear with **temp\_30** and **temp\_60** (Figure A2.2), supporting its use as the sole temperature predictor. There is a negative relationship between **mean\_elevation** and **yday**, and it is only weakly correlated with **temp\_90** ( $r = 0.31$ ). Flow variables (**flow\_30**, **flow\_60**, **flow\_90**) all show a decaying-exponential relationship with spawn date, with a strong year effect that suggests much of their explanatory signal is redundant with year as a fixed effect. Stream **slope** shows no meaningful relationship with spawn date.

## A2.2 Stream temperature (temp\_90)

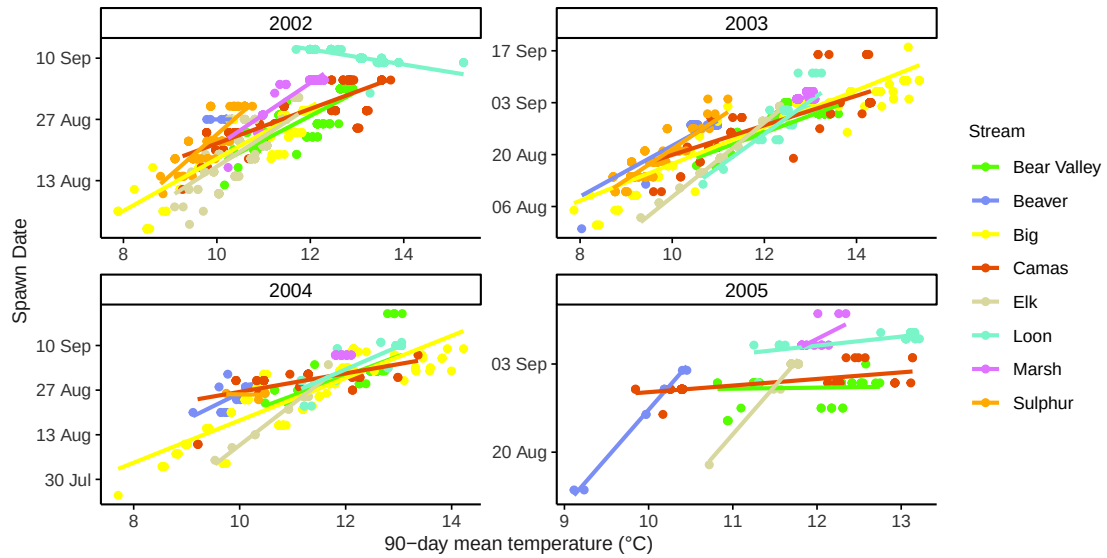


Figure A2.3: Relationship between 90 day antecedent stream temperature and spawn date by stream and year. Solid lines are linear fits.

### A2.3 mean\_elevation

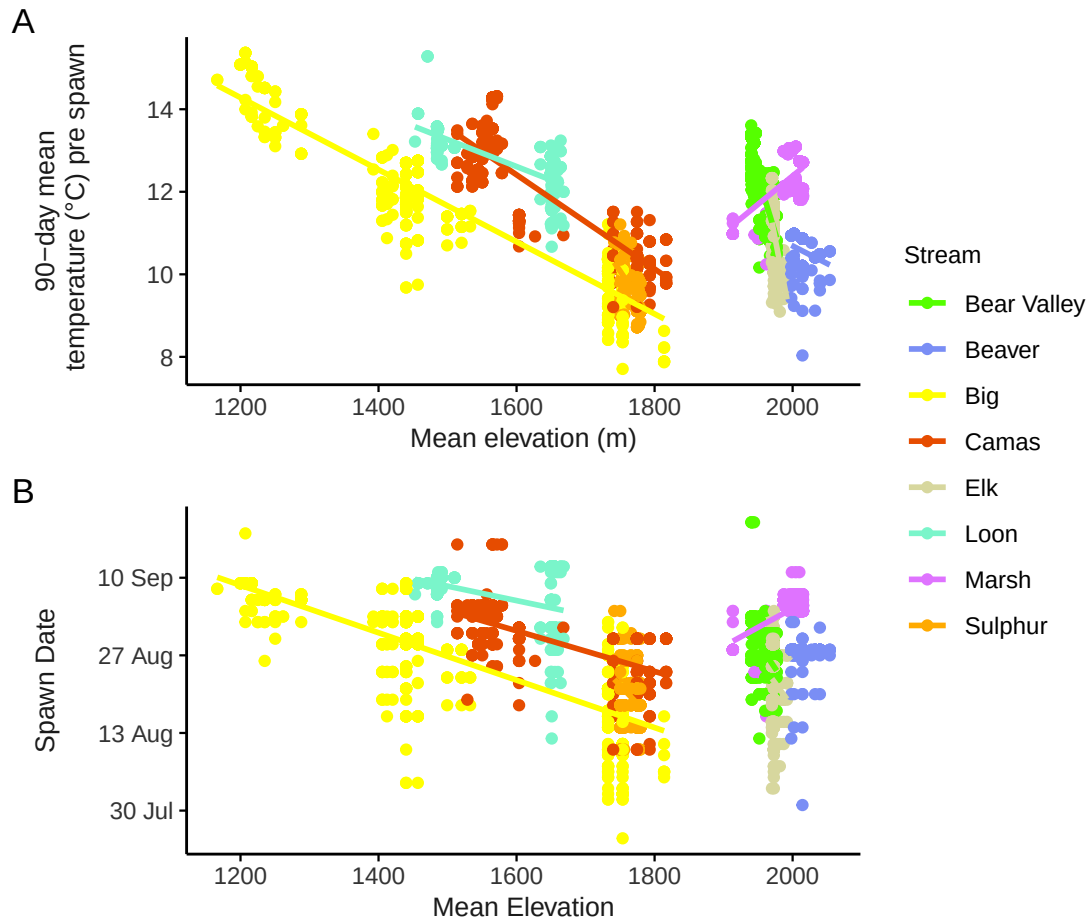


Figure A2.4: Relationship between (A) mean\_elevation and temp\_90 by stream and (B) mean\_elevation and spawn timing by stream and year. Solid lines are linear fits.

## A2.4 slope

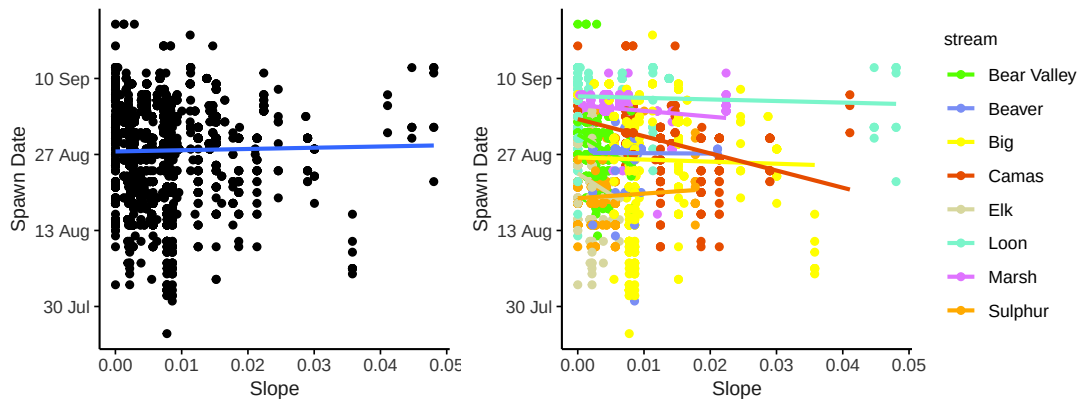


Figure A2.5: Bivariate relationship between `slope` and spawn date (A) and by stream (B). Solid lines are linear fits.

## A2.5 flow\_90

Annual streamflow conditions were considered because high-discharge years may impede upstream migration, delaying adult arrival on spawning grounds and pushing redd completion dates later in the season. We indexed flow using 30-, 60-, and 90-day averages prior to, during, and following the inferred spawn date at each redd location using discharge from a USGS gauge near the mouth of the Middle Fork (gauge 13309220). Because this gauge is downstream of all study reaches, the discharge index lacked spatial resolution across tributaries; moreover, year-to-year variation dominated its signal, producing strong collinearity with the year fixed effect ( $VIF > 1000$  in models including both), which precluded independent inference on flow conditions. Discharge was therefore excluded from further consideration

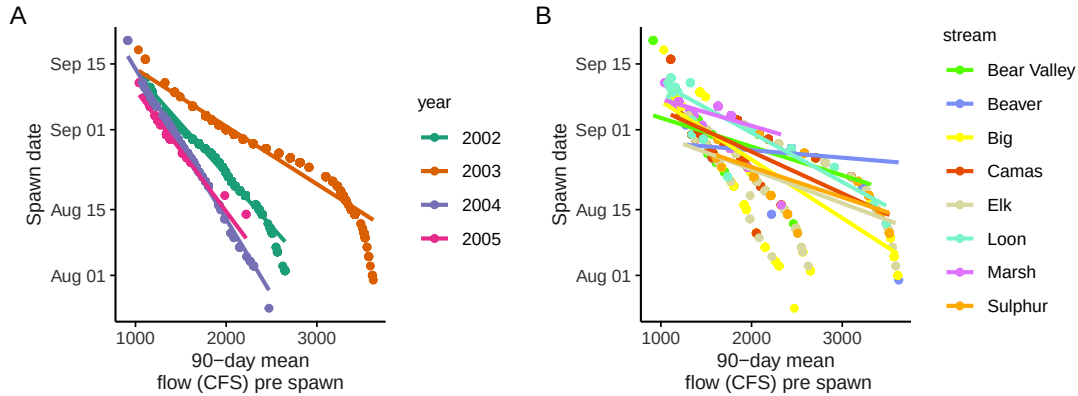


Figure A2.6: Spawn timing vs. 90-day mean discharge pre spawn at MF Lodge.

### A2.5.1 Why flow\_90 is problematic

Not spatially resolved

- We are modeling spawn timing at the redd level (COMID/stream)
- `flow_90` is calculated from a single downstream gauge
- This assumes flow conditions are identical across all sites
- Including it gives the illusion of spatially resolved variation that isn't there

Correlated with year

- `flow_90` varies mostly across years, strongly confounded with year
- flow-related signal is probably already captured by year fixed effect
- Including both `flow_90` and year risks collinearity

Spawn-time aligned flow does not equal experienced flow

- While `flow_90` is aligned to each redd's spawn date, it reflects a low basin gauge, not actual hydrologic conditions experienced at the redd site
- So it might be precisely wrong — aligned in time but irrelevant in space

## A3 Data Analysis

The final analysis dataset includes:

- **yday**: spawn date, continuous response variable
- **comid**, **stream**, **year**: grouping variables
- **temp\_90**: 90-day mean temperature pre-spawn, continuous variable

We used linear mixed-effects models to evaluate environmental predictors of Chinook salmon spawn timing, with redd observation day-of-year (**yday**) as the response variable. We scaled the continuous covariates to have a mean of 0 and standard deviation of 1 to assist convergence and interpretation. **COMID** will be included in all models as a random (intercept) effect to account for repeated measures on **COMIDs**. **Stream** will be treated as a fixed effect to compare average effects across streams, and **year** as a fixed effect to compare average effects across years. Further, based on exploratory analysis and biological expectations of nonlinear thermal responses, temperature effects were modeled using both linear and quadratic terms for the 90-day average stream temperature prior to spawning (**temp\_90** and  $I(\text{temp\_90}^2)$ ).

### A3.1 Initial Model Selection

We fit 31 additive models representing all combinations of fixed effects: **temp\_90**, **stream**, **year**, **slope**, and **mean\_elevation**. Model selection was performed using AIC, and model fit was evaluated using marginal and conditional  $R^2$ , RMSE, and intraclass correlation coefficients (ICC).

```
# one covariate
m1 <- lmer(yday ~ temp_90 + I(temp_90^2) + (1 | COMID), data = df_mod,
  REML = FALSE)
m2 <- lmer(yday ~ stream + (1 | COMID), data = df_mod, REML = FALSE)
m3 <- lmer(yday ~ year + (1 | COMID), data = df_mod, REML = FALSE)

# two covariates
m4 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + (1 | COMID), data = df_mod,
  REML = FALSE)
m5 <- lmer(yday ~ temp_90 + I(temp_90^2) + year + (1 | COMID), data = df_mod,
  REML = FALSE)
# m8 <- lmer(yday ~ temp_90 + I(temp_90^2) + mean_elevation +
# (1|COMID), data = df_mod, REML = FALSE) m9 <- lmer(yday ~ temp_90 +
# I(temp_90^2) + slope + (1|COMID), data = df_mod, REML = FALSE)
m6 <- lmer(yday ~ stream + year + (1 | COMID), data = df_mod, REML = FALSE)

# three covariates
m7 <- lmer(yday ~ temp_90 + I(temp_90^2) + stream + year + (1 | COMID),
  data = df_mod, REML = FALSE)
```

Table A3.1: AIC and model performance metrics for 7 additive candidate models.

| Model | df | AIC      | delta_AIC  | R2_marginal | R2_conditional | RMSE     | ICC       |
|-------|----|----------|------------|-------------|----------------|----------|-----------|
| m7    | 15 | 13473.12 | 0.00000    | 0.7293886   | 0.9800601      | 2.022117 | 0.9263154 |
| m5    | 8  | 13531.17 | 58.04838   | 0.6600787   | 0.9871140      | 2.021778 | 0.9620912 |
| m4    | 12 | 15807.98 | 2334.86171 | 0.7921275   | 0.8982595      | 3.110542 | 0.5105629 |
| m1    | 5  | 15848.81 | 2375.68889 | 0.7643117   | 0.9177407      | 3.108315 | 0.6509825 |



|    |    |          |            |           |           |          |           |
|----|----|----------|------------|-----------|-----------|----------|-----------|
| m6 | 13 | 18456.90 | 4983.78574 | 0.3741465 | 0.6557533 | 4.843560 | 0.4499565 |
| m3 | 6  | 18519.96 | 5046.84122 | 0.0144422 | 0.6647640 | 4.835146 | 0.6598515 |
| m2 | 10 | 18563.32 | 5090.19857 | 0.3631615 | 0.6425863 | 4.938363 | 0.4387687 |

The full additive model (m7) was the top-ranked model and was carried forward as the base model for subsequent evaluation of random slopes and temperature functional form.

## A3.2 Targeted model comparison

We compared our top additive model (m7) to a series of increasingly flexible models to evaluate the contribution of random slopes and temperature nonlinearity.

### A3.2.1 Random slopes for temp\_90

To account for potential variation in temperature sensitivity across sites, we extended m7 to include **COMID-specific random slopes for temp\_90**, yielding a model with the same fixed effects but with the random structure (1 + temp\_90 | COMID). Because the random effects structure differed, we fit both models using restricted maximum likelihood (REML).

```
m7_reml <- lmer(
  yday ~ temp_90 + I(temp_90^2) + stream + year + (1 | COMID),
  data = df_mod, REML = TRUE
)
m7_rs_reml <- lmer(
  yday ~ temp_90 + I(temp_90^2) + stream + year + (1 + temp_90 | COMID),
  data = df_mod, REML = TRUE
)
```

Table A3.2: Model selection for m7 vs. m7\\_rs.

| Model      | df | AIC      | delta_AIC | R2_marginal | R2_conditional | RMSE  | ICC   |
|------------|----|----------|-----------|-------------|----------------|-------|-------|
| m7_rs_reml | 17 | 12948.76 | 0.000     | 0.698       | 0.985          | 1.781 | 0.949 |
| m7_reml    | 15 | 13459.05 | 510.283   | 0.714       | 0.981          | 2.022 | 0.932 |

The random slope model (m7\_rs\_reml) substantially improved fit ( $\Delta\text{AIC} = 510$ ), reduced RMSE (2.02  $\rightarrow$  1.78 days), and slightly increased conditional  $R^2$  (0.981  $\rightarrow$  0.985). However, the marginal  $R^2$  declined from 0.714 to 0.698, reflecting a redistribution of explanatory power from fixed to random effects. This shift is expected when site-specific variation in temperature responses is modeled explicitly, and suggests that temperature sensitivity varies meaningfully among stream reaches. We next assessed residual behavior of m7\_rs\_reml to evaluate whether the quadratic temperature term adequately captured the nonlinear relationship before proceeding to functional form comparison.

## A3.3 Residual Diagnostics

DHARMA diagnostics for m7\_rs\_reml revealed adequate overall fit alongside a systematic pattern that motivated evaluation of a more flexible temperature functional form. The QQ plot showed close tracking of the diagonal despite a flagged KS test ( $p \approx 0$ ) and outlier test ( $p = 1 \times 10^{-5}$ ); with  $n = 3,016$ , DHARMA's

omnibus tests readily detect minor deviations that carry little practical significance, and the dispersion test was non-significant ( $p = 0.44$ ). More informative was the residuals vs. predicted plot, which showed a pronounced S-shaped pattern in the quantile smoothers — indicating mild non-uniformity across the prediction range, likely reflecting asymmetry in spawn timing at the distributional tails.

The residuals vs. temp\_90 plot was more diagnostic: quantile smoothers showed a clear monotonic decline across the temperature gradient, with residuals shifted consistently above 0.5 at cold sites and below 0.5 at warm sites. This systematic trend indicates the model was over-predicting spawn timing at low temperatures and under-predicting at high temperatures even after the quadratic term, suggesting the quadratic did not fully capture the nonlinear temperature–spawn timing relationship. Residuals plotted against stream and year were also flagged (Levene test significant), reflecting heterogeneity not fully absorbed by the fixed effects: Elk and Marsh creeks showed compressed distributions shifted below 0.5, while Big and Camas were shifted high, and 2005 exhibited notably lower residuals than other years.

Together, these patterns motivated evaluation of a more flexible functional form for the temperature effect.

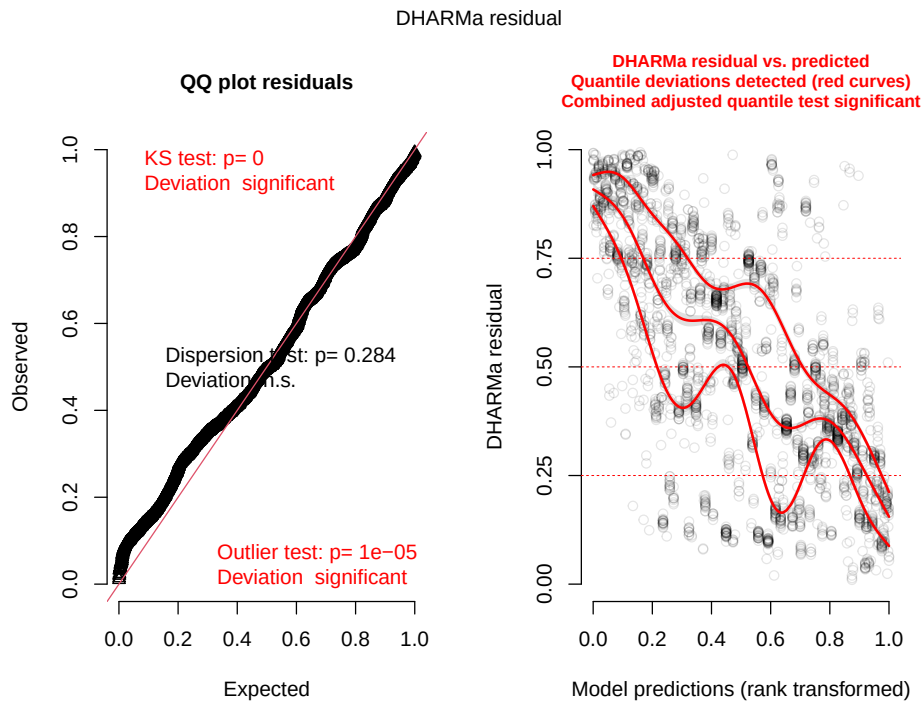


Figure A3.1: DHARMa residuals 1.

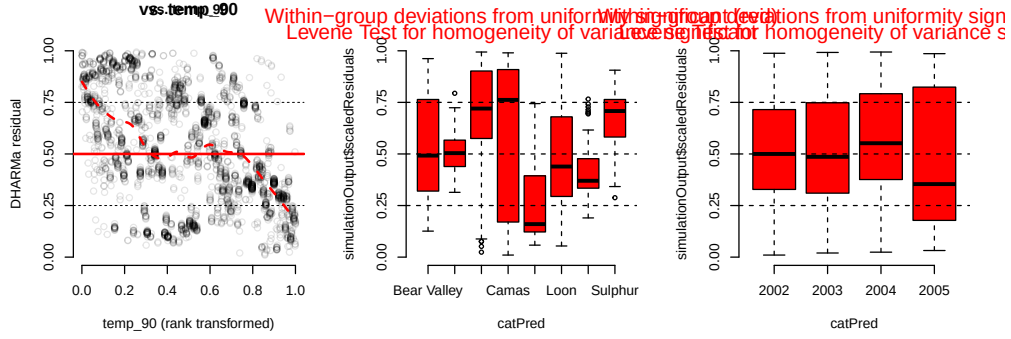


Figure A3.2: DHARMA residuals 2.

### A3.4 Evaluation of Temperature Functional Form

The  $df=3$  spline substantially outperformed the quadratic ( $\Delta AIC = 139$ ) with lower RMSE and improved residual behavior across the temperature gradient. The  $df=4$  spline showed further AIC improvement ( $\Delta AIC = 66$  over  $df=3$ ) but produced biologically implausible predictions at warm-end temperature extremes — spawn timing decreasing sharply above  $\sim 13^{\circ}\text{C}$  — indicative of overfitting. The  $df=3$  spline was therefore selected as the final temperature functional form.

Table A3.3: AIC comparison of quadratic vs. spline forms.

| Model       | df | AIC      | delta     |
|-------------|----|----------|-----------|
| mod_spline4 | 19 | 12761.28 | 0.00000   |
| mod_spline3 | 18 | 12827.09 | 65.81274  |
| m7_rs       | 17 | 12965.97 | 204.69238 |

Table A3.4: Model performance for quadratic vs. spline forms

| Name        | R2_conditional | R2_marginal | ICC       | RMSE     | Performance_Score |
|-------------|----------------|-------------|-----------|----------|-------------------|
| mod_spline4 | 0.9950744      | 0.4831279   | 0.9904704 | 1.666518 | 0.8333333         |
| mod_spline3 | 0.9903689      | 0.6087398   | 0.9753845 | 1.715325 | 0.3942136         |
| m7_rs       | 0.9840995      | 0.7132881   | 0.9445417 | 1.781828 | 0.1666667         |

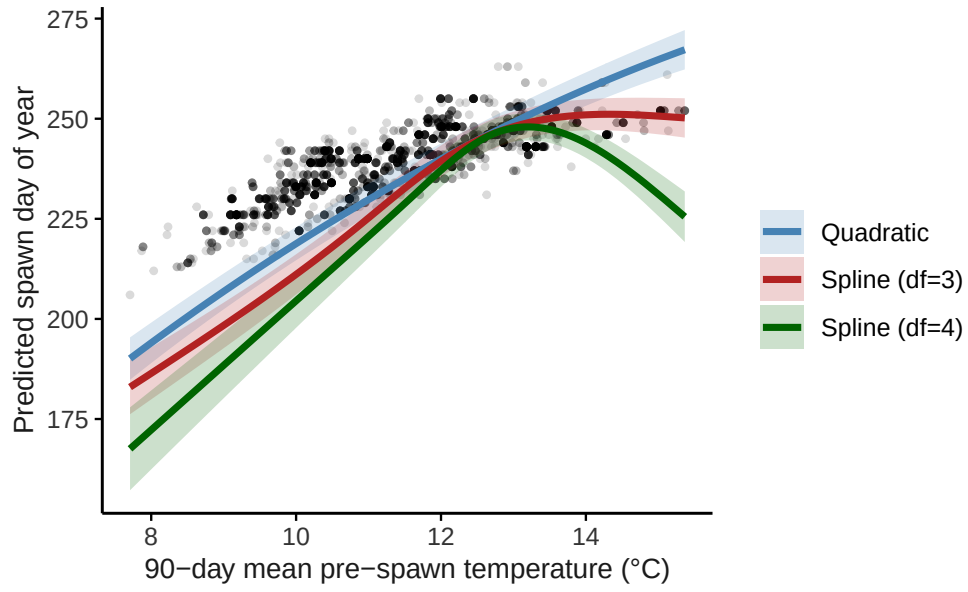


Figure A3.3: Prediction comparison plots.

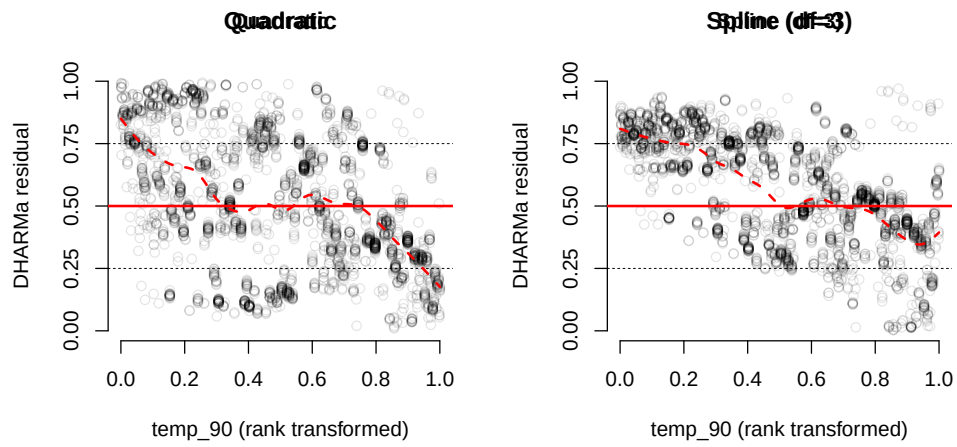


Figure A3.4

## A4 Final Model Diagnostics and Interpretation

```
mod_final <- lmer(  
  yday ~ ns(temp_90, df = 3) + stream + year + (1 + temp_90 | COMID),  
  data = df_mod, REML = TRUE  
)
```

### A4.1 Residual diagnostics

Diagnostics for the final model (`mod_final`) indicated strong overall performance alongside several patterns that warrant transparent reporting (Figure A4.1, Figure A4.2), Figure A4.3).

The posterior predictive check showed excellent agreement between observed and model-predicted distributions of spawn timing, with nearly overlapping density curves and no systematic deviation in the central tendency or spread. Collinearity was low, with variance inflation factors (VIFs) for all fixed effects well below 5. Influential observations were present — several redds showed moderate-to-high leverage — but standardized residuals remained within acceptable bounds and no observations were extreme enough to justify removal.

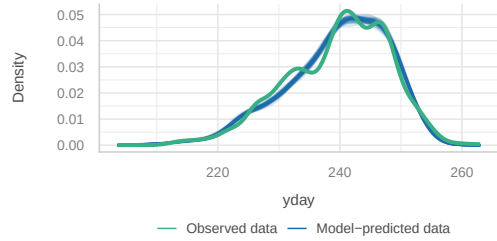
The residual vs. fitted and homogeneity of variance plots showed mild heteroscedasticity, with slight funneling at lower fitted values. This pattern likely reflects the skewed distribution of early spawn dates rather than model misspecification, and was insufficient to invalidate inference. The normality of residuals Q-Q plot showed more pronounced tail deviations — particularly in the upper tail — suggesting the Gaussian error assumption is imperfect. Random effect Q-Q plots for both intercepts and slopes tracked the diagonal reasonably well with modest tail departures, indicating acceptable distributional assumptions for the mixed-effects structure.

DHARMA simulated residuals corroborated these patterns while providing additional diagnostic nuance. The KS test ( $p \approx 0$ ), dispersion test ( $p = 0.004$ ), and outlier test ( $p = 1 \times 10^{-5}$ ) were all flagged significant; however, with  $n = 3,016$ , DHARMA’s omnibus tests are highly sensitive and will reliably detect minor deviations that carry little inferential consequence. More informative were the directional patterns: the residuals vs. predicted plot showed a residual S-shaped pattern in the quantile smoothers, and residuals plotted against stream and year remained flagged, indicating that the fixed effects for these grouping variables do not fully account for distributional heterogeneity across sites and years. Importantly, the residuals vs. temp\_90 plot showed a substantially flatter quantile trend compared to the quadratic model, confirming that the natural spline captured the nonlinear temperature–spawn timing relationship more faithfully.

Taken together, these diagnostics indicate that the final model is well-specified for its primary inferential purpose — characterizing the population-level temperature–spawn timing relationship and its spatial heterogeneity — while acknowledging that the Gaussian error structure imperfectly represents the distributional properties of spawn dates. Crucially, the persistent residual structure across streams and years is not a modeling failure but an ecological signal: it reflects fine-scale habitat heterogeneity — groundwater connectivity, hyporheic exchange, local geomorphology, run composition — that operates below the resolution of available covariates and cannot be absorbed by stream identity or year alone. This is precisely the unmeasured variation that the random slope structure was designed to partially accommodate, and its persistence in the residuals reinforces the central finding of this paper: that spawn timing diversity in the Middle Fork is sustained by reach-scale processes that temperature measurements and fixed grouping variables, however carefully assembled, cannot fully characterize.

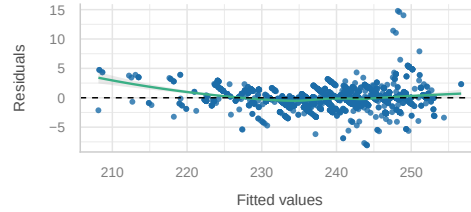
### Posterior Predictive Check

Model-predicted lines should resemble observed data line



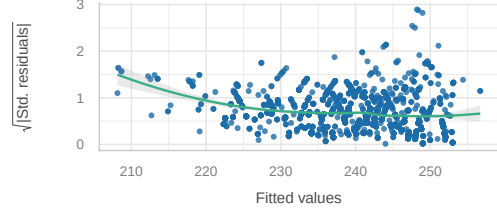
### Linearity

Reference line should be flat and horizontal



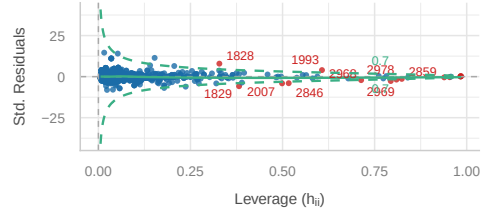
### Homogeneity of Variance

Reference line should be flat and horizontal



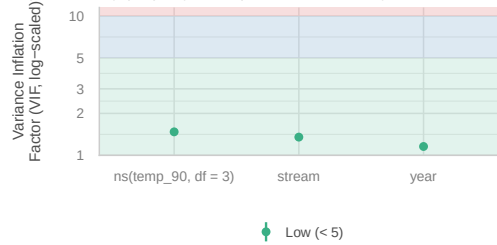
### Influential Observations

Points should be inside the contour lines



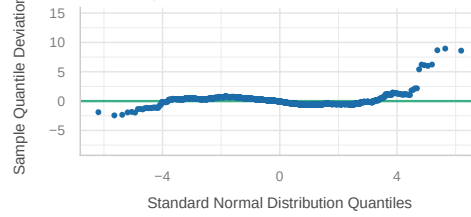
### Collinearity

High collinearity (VIF) may inflate parameter uncertainty



### Normality of Residuals

Dots should fall along the line



### Normality of Random Effects (COMID)

Dots should be plotted along the line

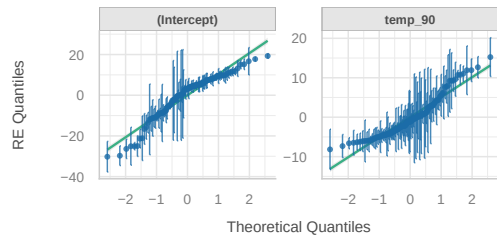


Figure A4.1: Residual diagnostics for mod\_final.

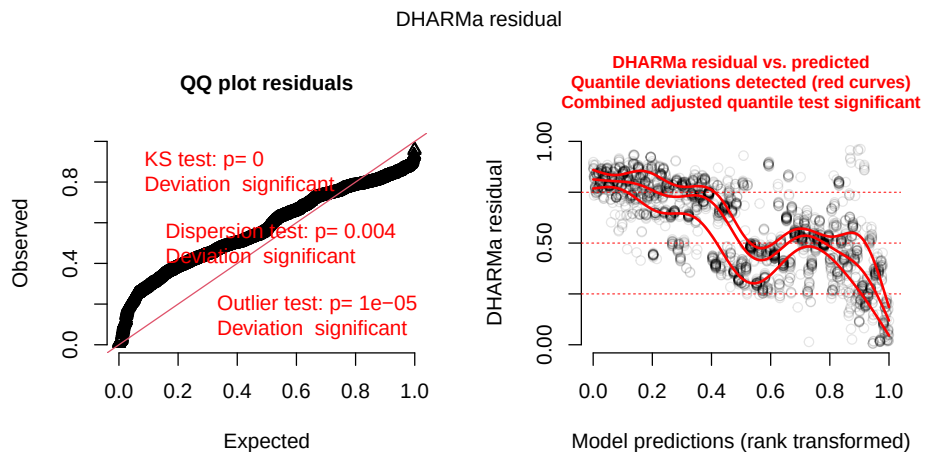


Figure A4.2: DHARMa residuals 1.

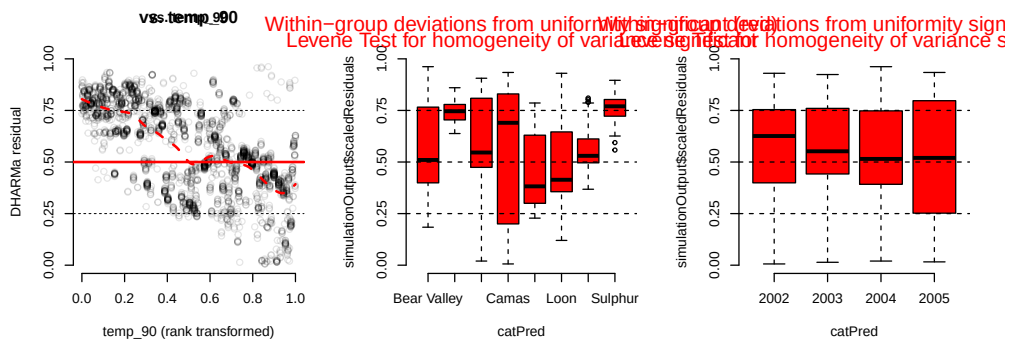


Figure A4.3: DHARMa residuals 2.

## A4.2 Model parameters and performance

Table A4.1: Model selection for linear mixed-effects models of Chinook salmon spawn timing.

| Model                                             | $\Delta\text{AIC}$ | RMSE | $R^2$<br>(marg.) | $R^2$<br>(cond.) |
|---------------------------------------------------|--------------------|------|------------------|------------------|
| <b>1. Additive models (random intercept only)</b> |                    |      |                  |                  |
| m7: temp + stream + year                          | 0.0                | 2.02 | 0.729            | 0.980            |
| m5: temp + year                                   | 58.1               | 2.02 | 0.660            | 0.987            |
| m4: temp + stream                                 | 2334.9             | 3.11 | 0.792            | 0.898            |
| m1: temp                                          | 2375.7             | 3.11 | 0.764            | 0.918            |
| m6: stream + year                                 | 4983.8             | 4.84 | 0.374            | 0.656            |
| m3: year                                          | 5046.9             | 4.84 | 0.014            | 0.665            |
| m2: stream                                        | 5090.2             | 4.94 | 0.363            | 0.643            |
| <b>2. Random slope evaluation (REML fit)</b>      |                    |      |                  |                  |
| m7_rs: + random slopes for temp                   | 0.0                | 1.78 | 0.698            | 0.985            |
| m7: random intercept only                         | 510.2              | 2.02 | 0.714            | 0.981            |
| <b>3. Temperature functional form (ML fit)</b>    |                    |      |                  |                  |
| Natural spline df = 3                             | 0.0                | 1.72 | 0.609            | 0.990            |
| Quadratic: temp + temp <sup>2</sup>               | 138.9              | 1.78 | 0.713            | 0.984            |
| Natural spline df = 4 <sup>†</sup>                | -65.8              | 1.67 | 0.483            | 0.995            |

Table A4.2: Parameter estimates for the final linear mixed-effects model of Chinook salmon spawn timing.

| Parameter                               | Est.    | 95% CI           | <i>p</i> |
|-----------------------------------------|---------|------------------|----------|
| <b>Fixed effects</b>                    |         |                  |          |
| Intercept                               | 183.11  | [176.16, 190.05] | <0.001   |
| Temp. spline term 1                     | 59.61   | [55.29, 63.93]   | <0.001   |
| Temp. spline term 2                     | 94.54   | [85.83, 103.25]  | <0.001   |
| Temp. spline term 3                     | 52.14   | [46.22, 58.06]   | <0.001   |
| Stream: Beaver                          | 14.88   | [7.75, 22.02]    | <0.001   |
| Stream: Big                             | 3.84    | [-1.11, 8.80]    | 0.128    |
| Stream: Camas                           | 5.53    | [0.28, 10.77]    | 0.039    |
| Stream: Elk                             | 12.23   | [5.08, 19.37]    | <0.001   |
| Stream: Loon                            | 4.15    | [-0.86, 9.16]    | 0.104    |
| Stream: Marsh                           | 5.25    | [-0.34, 10.84]   | 0.065    |
| Stream: Sulphur                         | 13.44   | [7.02, 19.85]    | <0.001   |
| Year: 2003                              | -4.83   | [-5.05, -4.60]   | <0.001   |
| Year: 2004                              | 2.76    | [2.53, 2.99]     | <0.001   |
| Year: 2005                              | 3.72    | [3.42, 4.01]     | <0.001   |
| <b>Random effects</b>                   |         |                  |          |
| $\sigma^2$ (residual)                   | 3.12    | NA               | NA       |
| $\tau_{00}$ COMID (intercept)           | 128.20  | NA               | NA       |
| $\tau_{11}$ COMID (slope)               | 33.44   | NA               | NA       |
| $\rho_{01}$ COMID (int. $\times$ slope) | -0.83   | NA               | NA       |
| ICC                                     | 0.98    | NA               | NA       |
| N (COMID)                               | 104.00  | NA               | NA       |
| N (obs.)                                | 3016.00 | NA               | NA       |
| <b>Model performance</b>                |         |                  |          |
| $R^2$ marginal                          | 0.60    | NA               | NA       |
| $R^2$ conditional                       | 0.99    | NA               | NA       |
| RMSE (days)                             | 1.71    | NA               | NA       |



#### A4.2.1 Predictions against original data

Model predictions closely matched observed spawn timing, with predicted values aligning well along the 1:1 line (Figure A4.4), consistent with the strong conditional  $R^2$  reported above.

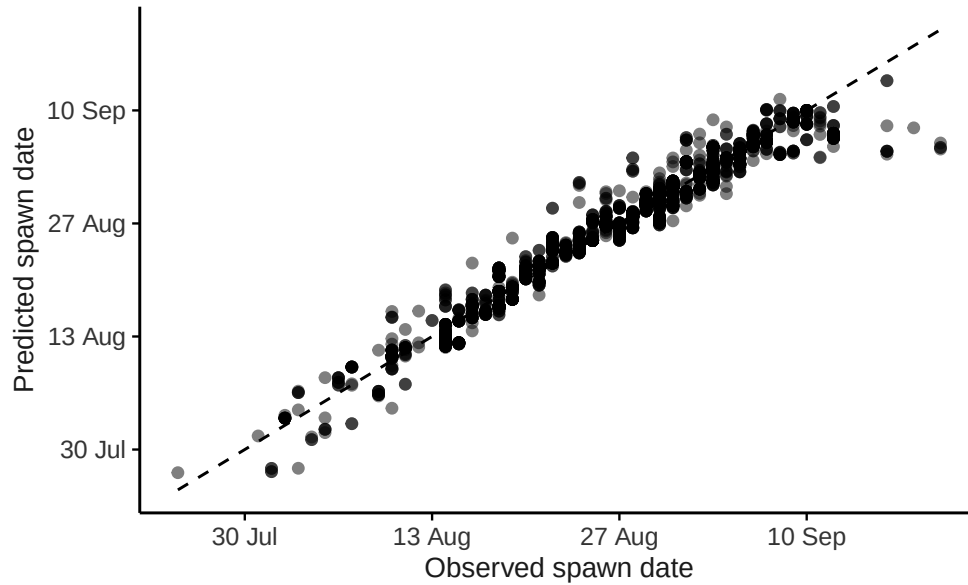


Figure A4.4: Predicted vs. observed spawn timing for Chinook salmon across all years and streams. The dashed line shows the 1:1 line. Points represent individual redd observations.

### A4.3 Temperature effects

Predicted spawn timing increased nonlinearly with 90-day mean pre-spawn temperature across all streams and years, leveling off at high temperatures consistent with biological expectations (Figure A4.5). Stream-to-stream variation reflects both fixed stream effects and COMID-specific random intercepts and slopes. Year-specific offsets were modest, with curvature consistent across years.

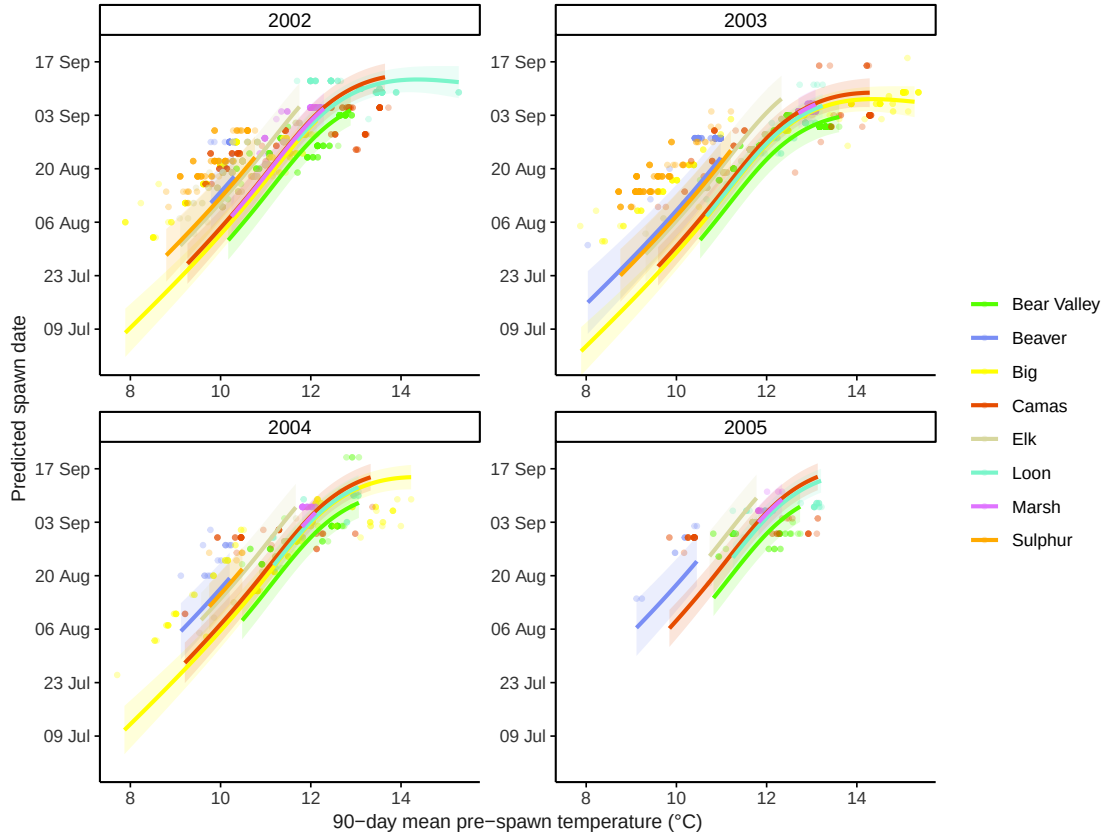


Figure A4.5: Predicted relationship between spawn timing and 90-day pre-spawn mean temperature by stream and year. Lines represent population-level model predictions from the final mixed-effects model. Colored points illustrate observed spawn timing; shaded ribbons represent 95% confidence intervals.

## A4.4 Stream and year effects

Marginal mean spawn timing was estimated at each factor level, averaging over random effects to characterize population-level patterns.

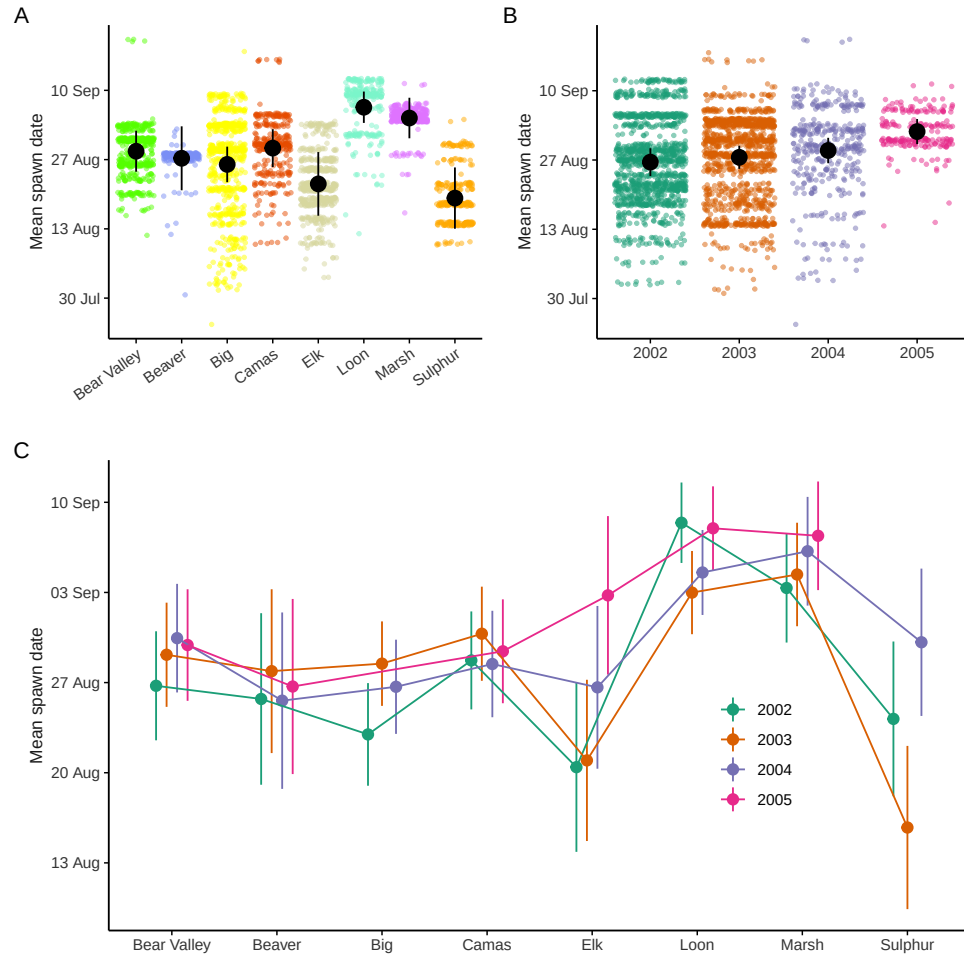


Figure A4.6: Predicted mean spawn dates by stream (A), year (B), and stream  $\times$  year (C) from the final mixed-effects model. Black points and lines in panels A and B represent estimated marginal means and 95% confidence intervals averaged over random effects. Colored points show individual redd observations. Colored points and lines in panel C represent stream-specific marginal means and 95% confidence intervals by year.

## Stream contrasts

Significant differences in spawn timing were observed between many stream pairs (Table A4.3), particularly involving Loon (later spawning) and Sulphur (earlier spawning). For example, fish in Loon spawned significantly later than in Bear Valley, Camas, and Elk, while Sulphur exhibited significantly earlier timing than all other streams except Elk. These patterns reflect spatial heterogeneity in temperature and elevation across streams that is not fully captured by fixed effects alone.

Table A4.3: Pairwise contrasts among stream effects on predicted spawn timing. Contrasts represent estimated differences in predicted spawn day of year between streams from the final model. Positive values indicate later predicted spawn timing in the first stream compared to the second. P-values are uncorrected.

| Level1  | Level2      | Difference | SE   | CI_low | CI_high | t     | df   | p    |
|---------|-------------|------------|------|--------|---------|-------|------|------|
| Beaver  | Bear Valley | -1.43      | 3.68 | -8.65  | 5.80    | -0.39 | 2998 | 0.70 |
| Big     | Bear Valley | -2.66      | 2.54 | -7.65  | 2.33    | -1.05 | 2998 | 0.30 |
| Camas   | Bear Valley | 0.63       | 2.68 | -4.62  | 5.88    | 0.24  | 2998 | 0.81 |
| Elk     | Bear Valley | -6.60      | 3.68 | -13.83 | 0.62    | -1.79 | 2998 | 0.07 |
| Loon    | Bear Valley | 8.90       | 2.57 | 3.86   | 13.94   | 3.46  | 2998 | 0.00 |
| Marsh   | Bear Valley | 6.72       | 2.86 | 1.11   | 12.32   | 2.35  | 2998 | 0.02 |
| Sulphur | Bear Valley | -9.47      | 3.51 | -16.35 | -2.59   | -2.70 | 2998 | 0.01 |
| Big     | Beaver      | -1.23      | 3.37 | -7.85  | 5.38    | -0.37 | 2998 | 0.72 |
| Camas   | Beaver      | 2.06       | 3.50 | -4.80  | 8.92    | 0.59  | 2998 | 0.56 |
| Elk     | Beaver      | -5.18      | 4.17 | -13.35 | 3.00    | -1.24 | 2998 | 0.21 |
| Loon    | Beaver      | 10.33      | 3.57 | 3.33   | 17.32   | 2.89  | 2998 | 0.00 |
| Marsh   | Beaver      | 8.14       | 3.75 | 0.78   | 15.50   | 2.17  | 2998 | 0.03 |
| Sulphur | Beaver      | -8.04      | 3.80 | -15.49 | -0.60   | -2.12 | 2998 | 0.03 |
| Camas   | Big         | 3.29       | 2.34 | -1.31  | 7.89    | 1.40  | 2998 | 0.16 |
| Elk     | Big         | -3.94      | 3.39 | -10.59 | 2.70    | -1.16 | 2998 | 0.24 |
| Loon    | Big         | 11.56      | 2.30 | 7.05   | 16.07   | 5.02  | 2998 | 0.00 |
| Marsh   | Big         | 9.38       | 2.60 | 4.27   | 14.48   | 3.60  | 2998 | 0.00 |
| Sulphur | Big         | -6.81      | 3.08 | -12.86 | -0.76   | -2.21 | 2998 | 0.03 |
| Elk     | Camas       | -7.23      | 3.52 | -14.13 | -0.34   | -2.06 | 2998 | 0.04 |
| Loon    | Camas       | 8.27       | 2.43 | 3.51   | 13.03   | 3.41  | 2998 | 0.00 |
| Marsh   | Camas       | 6.09       | 2.72 | 0.75   | 11.43   | 2.23  | 2998 | 0.03 |
| Sulphur | Camas       | -10.10     | 3.27 | -16.50 | -3.70   | -3.09 | 2998 | 0.00 |
| Loon    | Elk         | 15.50      | 3.56 | 8.53   | 22.47   | 4.36  | 2998 | 0.00 |
| Marsh   | Elk         | 13.32      | 3.75 | 5.97   | 20.67   | 3.55  | 2998 | 0.00 |
| Sulphur | Elk         | -2.87      | 3.88 | -10.48 | 4.75    | -0.74 | 2998 | 0.46 |
| Marsh   | Loon        | -2.18      | 2.58 | -7.23  | 2.87    | -0.85 | 2998 | 0.40 |
| Sulphur | Loon        | -18.37     | 3.41 | -25.06 | -11.68  | -5.39 | 2998 | 0.00 |
| Sulphur | Marsh       | -16.19     | 3.60 | -23.24 | -9.13   | -4.50 | 2998 | 0.00 |

### Year contrasts

There was a clear trend toward later spawning over the four-year period (Table A4.4). Spawning in 2005 occurred significantly later than in all previous years. Differences between 2002 and 2003 were not statistically significant, but later years (2004 and especially 2005) were associated with a progressive delay in mean spawn timing. This temporal shift likely reflects interannual variability in temperature and flow conditions.

Table A4.4: Pairwise contrasts among year effects on predicted spawn timing. Contrasts represent estimated differences in predicted spawn day of year between years from the final model. Positive values indicate later spawning in the first year compared to the second. P-values are uncorrected.

| Level1 | Level2 | Difference | SE   | CI_low | CI_high | t    | df   | p    |
|--------|--------|------------|------|--------|---------|------|------|------|
| 2003   | 2002   | 0.97       | 0.61 | -0.23  | 2.18    | 1.59 | 2998 | 0.11 |
| 2004   | 2002   | 2.35       | 0.67 | 1.03   | 3.67    | 3.48 | 2998 | 0.00 |
| 2005   | 2002   | 6.18       | 0.64 | 4.92   | 7.43    | 9.64 | 2998 | 0.00 |
| 2004   | 2003   | 1.37       | 0.66 | 0.08   | 2.67    | 2.08 | 2998 | 0.04 |
| 2005   | 2003   | 5.20       | 0.75 | 3.74   | 6.66    | 6.98 | 2998 | 0.00 |
| 2005   | 2004   | 3.83       | 0.84 | 2.17   | 5.48    | 4.53 | 2998 | 0.00 |

### A4.5 Reach-level variation (random effects)

Random intercepts and slopes varied considerably among COMIDs, with substantial spread both within and between streams (Figure A4.7 A-B). Random intercepts represent reach-level deviations in baseline spawn timing after accounting for fixed effects: a positive intercept indicates a reach where salmon spawn later than expected given temperature, stream identity, and year; a negative intercept indicates earlier-than-expected spawning. Random slopes represent deviations in thermal sensitivity: steeper slopes indicate stronger temperature dependence, shallower slopes reflect weaker or buffered responses. Among-reach variance was substantial for both baseline timing (128.20) and thermal sensitivity (33.44), and the pattern of variation differed among streams.

The intercept-slope correlation was strongly negative ( $-0.83$ ; Figure A4.7 C), indicating that reaches with earlier baseline spawn timing exhibited substantially greater thermal sensitivity. Early-spawning, likely higher-elevation reaches are not only earlier — they are also more responsive to temperature variation, suggesting greater phenotypic plasticity. In contrast, later-spawning, lower-elevation reaches exhibit shallower thermal responses, possibly reflecting operation near thermal thresholds or stronger constraint by other cues such as photoperiod or migration timing.

Figure A4.7 D translates these random effect patterns into predicted temperature–spawn timing relationships for individual COMIDs, illustrating how the combination of varying intercepts and slopes generates a diverse family of reach-specific trajectories around the population mean. While the overall relationship between pre-spawn temperature and spawn timing is positive and nonlinear, the strength and shape of this response varies substantially among reaches, likely reflecting fine-scale differences in hydrology, geomorphology, and local thermal buffering capacity. This reach-level heterogeneity is the mechanism by which fine-scale habitat diversity sustains diverse spawning portfolios within the Middle Fork.

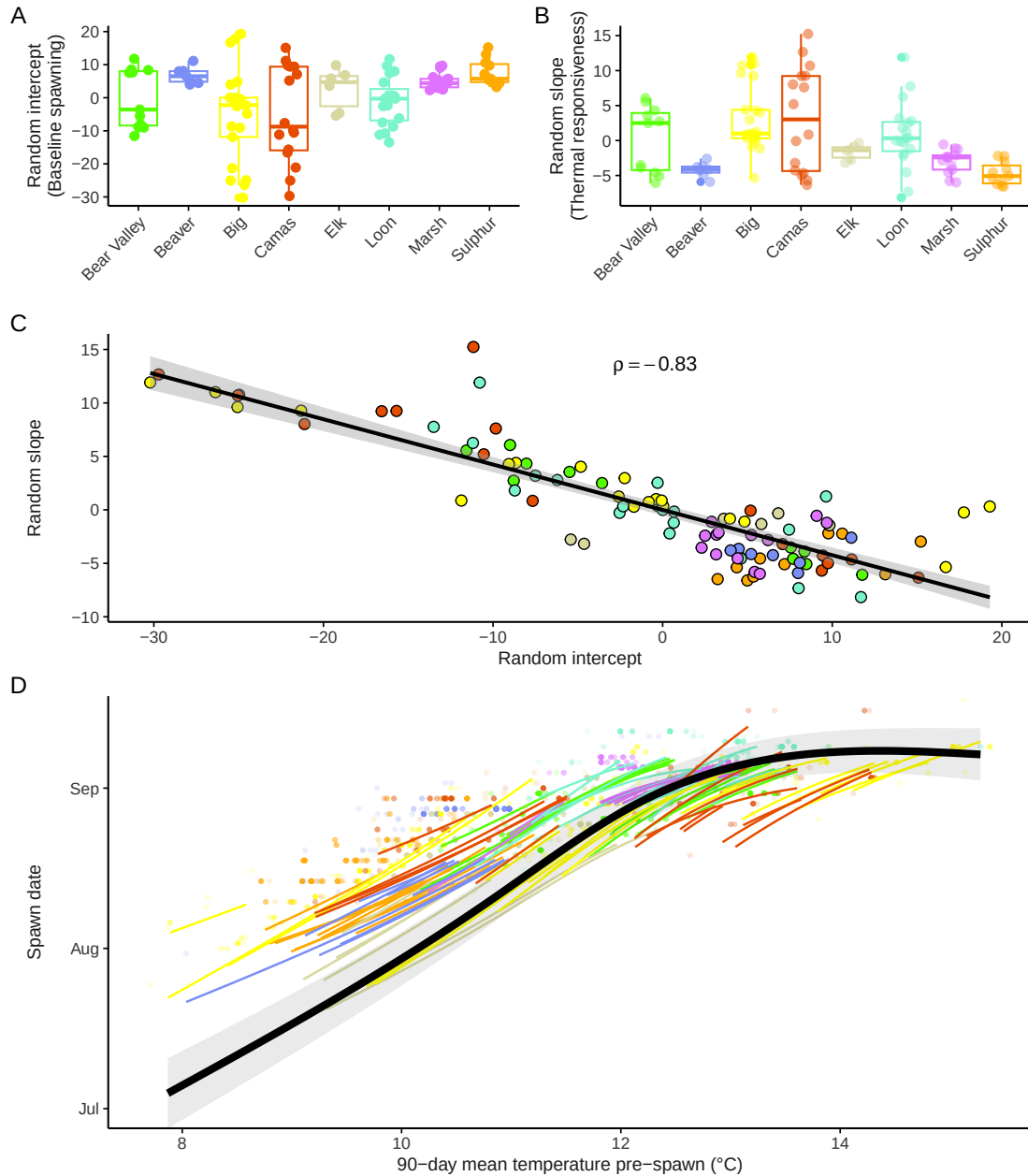


Figure A4.7: Reach-level (COMID) random effects and predicted spawn timing relationships from the final mixed-effects model. (A) Random intercepts by stream, representing reach-level deviations in baseline spawn timing after accounting for fixed effects; positive values indicate later-than-expected spawning. (B) Random slopes by stream, representing reach-level deviations in thermal sensitivity; larger positive values indicate stronger temperature dependence. Points in A and B represent individual COMIDs. (C) Correlation between random intercepts and slopes across all COMIDs, colored by stream; the strong negative relationship ( $\rho = -0.83$ ) indicates that reaches with earlier baseline spawn timing exhibit greater thermal sensitivity. (D) Predicted temperature-spawn timing relationships for individual COMIDs (colored lines) and the population mean response (black line with 95% confidence interval).